INCH-POUND
MS35340G
13 June 2022
SUPERSEDING
MS35340F
13 December 1993

DETAIL SPECIFICATION SHEET

WASHER, LOCK-HELICAL SPRING, EXTRA DUTY (EXTRA-HEAVY) SERIES

Reactivated after 13 June 2022 and may be used for new and existing designs and acquisitions.

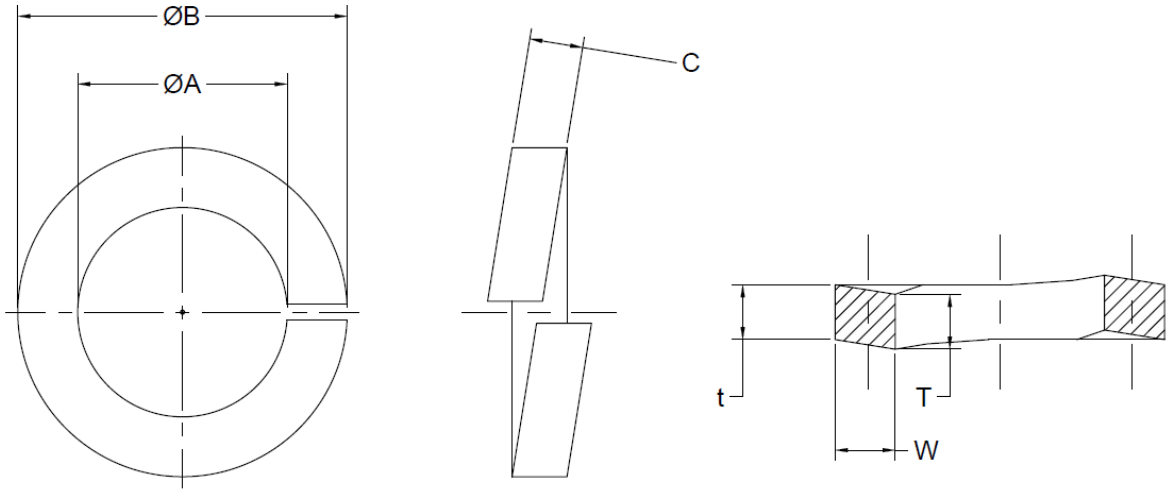


FIGURE 1. Washer, lock-helical spring.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Dash numbers and dimensions (inch).

NOMINAL SIZE		DASH NUMBER /1, /2	DASH NUMBER /1	ØA INSIDE DIAMETER		ØB OUTSIDE DIAMETER	WASHER SECTION	
							W WIDTH	C THICKNESS $\frac{T+t}{2}$
				MIN	MAX	MAX	MIN	-0.000
NO.2	0.086	39	39MZ	Dimensions per ASME B18.21.1				
NO.4	0.112	40	40MZ					
NO.6	0.138	41	41MZ					
NO.8	0.164	42	42MZ					
NO.10	0.190	43	43MZ					
1/4	0.250	44	44MZ					
5/16	0.312	45	45MZ					
3/8	0.375	46	46MZ					
7/16	0.437	47						
1/2	0.500	48	48MZ					
9/16	0.562	49						
5/8	0.625	50	50MZ					
3/4	0.750	51	51MZ					
7/8	0.875	52						
1	1.000	53	53MZ					
1-1/8	1.125	54						
1-1/4	1.250	55	55MZ					
1-3/8	1.375	56						
1-1/2	1.500	57	57MZ					

/1 See REQUIREMENT 3.
/2 Inactive for new design.

REQUIREMENTS:

1 Material. Washers shall be made from steel, carbon, SAE 1050 to SAE 1080 (UNS G10500 to G10800), (see SAE J403).

2 Hardness. Washers shall be hardened to HRC 38 – 46, or equivalent.

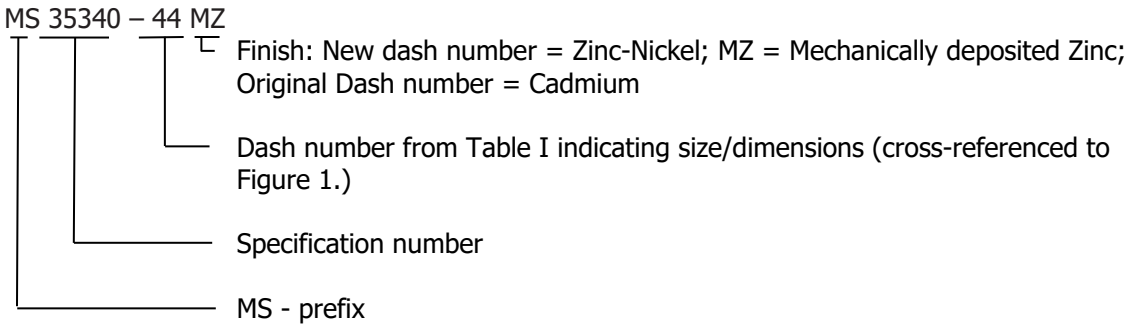
3 Protective finish. Mechanically deposited zinc (Dash Number plus MZ) in accordance with ASTM B695, Type II, except the supplementary treatment shall be hexavalent chromium-free
Cadmium [1/] plate in accordance with SAE AMS-QQ-P-416 (Original Dash number, Inactive for new design)

[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List.

4 Conformance. Certificates of compliance to paragraphs 1, 2, and 3 shall be provided.

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5 Part identifying number. The PIN consists of the following:



Example: MS35340-44MZ is a washer, lock-helical spring, extra heavy duty, mechanically deposited zinc, having a nominal size of 0.250.

6 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids. Copies of MIL specifications are available online at https://quicksearch.dla.mil. Copies of ASTM specifications are available online at https://www.astm.org/Standard/standards-and-publications. Copies of ASME specifications are available online at https://www.asme.org/codes-standards/find-codes-standards. Copies of SAE specifications are available online at https://www.sae.org/standards.

7 Design. For design feature purposes, this document takes precedence over all other documents.

8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Washers, lock-helical spring, extra duty series per ASME B18.21.1 may be described per ASME B18.24 (2020) PIN code using prefix "AEW09". However, a hexavalent chromium-free mechanical zinc with passivation is not yet available as a finish code.

NOTE: All PIN MS35340 washers having the original part numbers are inactive for new design after 13 June 2022. Use only mechanically deposited Zinc coated washers for new design; these may be used for sustainment.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 99
DLA – GS

Preparing activity:
Army – AT

(Project 5310-2022-017)

Review activities:
Army – AR, AV, CR4, GL, MI
Navy – OS, SH, YD

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